MD. ARMAN HASSAN

B.Sc. in Mechanical Engineering, BUET

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SUMMARY

Mechanical Engineering graduate from Bangladesh University of Engineering and Technology (BUET) with a CGPA of 3.88/4.00 and consistent placement on the Dean's List. Experienced in molecular dynamics research, with publications and ongoing research projects, alongside professional experience in mechanical design and manufacturing. Also experienced in student training, mentorship, and technical team leadership, with a strong interest in teaching.

EDUCATION

B.Sc. in Mechanical Engineering **June 2026**
Bangladesh University of Engineering and Technology
- CGPA: 3.88/4.00 | 12th out of 186 students

Higher Secondary Certificate (HSC) **2020**
Notre Dame College
- GPA: 5.00/5.00

Secondary School Certificate (SSC) **2018**
Birshreshtha Noor Mohammad Public College
- GPA: 5.00/5.00

UNDERGRADUATE THESIS

Molecular Dynamics Simulation of PDMS–hBN Nanocomposites for Thermal Conductivity Enhancement
Supervisor: Dr. Md. Ashiqur Rahman, Professor, Department of Mechanical Engineering, BUET

- Investigated thermal transport in PDMS–hBN nanocomposites using molecular dynamics simulations in LAMMPS, evaluating the effect of hBN filler fraction in aligned laminate configurations.
- Studied the effects of carbon-doped hBN and graphene/hBN heterostructure fillers on composite thermal conductivity at fixed filler loading.

PUBLICATIONS

Design and Fabrication of a Versatile and Cost-Effective Myoelectric Prosthetic Wrist
Md. Arman Hassan, Tanjim Azad Jarif, Jubaer Al-Amin Emon, Lamia Tasnim Suhi
8th International Conference on Engineering Research, Innovation and Education (ICERIE 2025), SUST
Atlantis Press (*Advances in Engineering Research*, pp. 598–606)
DOI: 10.2991/978-94-6463-884-4_72

- Analyzed forearm EMG signals and developed a threshold-based control algorithm using signal averages and standard deviations to enable intuitive grasp and release motions.
- Designed compliant TPU fingers with integrated living hinges, exploiting material flexibility and elasticity to achieve natural, adaptive grasping without conventional mechanical finger joints.

Design and Performance Optimization of a Multi-Purpose Autonomous Rover for Planetary and Terrestrial Applications
Sahib U. Rauf, Safwan Sakib, **Md. A. Hassan**, Sk. F. Sabik, Nahian P. Jaman, Md. Arafath R. Nishat, Tawhidul I. Tahsif, Nafisa A. Tanisha, Salman R. Sunny, Kazi A. Rahman
15th International Conference on Mechanical Engineering (ICME 2025), BUET
Published on SSRN
DOI: 10.2139/ssrn.6202378

- Contributed to the mechanical design and development of the modular Prochesta V3.0 autonomous rover, including its mobility and robotic manipulation systems.
- Developed a compact 5-DOF robotic arm using stainless steel, aluminum, and 3D-printed components, balancing payload capacity, reach, manufacturability, and weight.

PROJECTS

Molecular Dynamics Study of CO_2 Diffusion Slowdown in MOF Pores Under N_2 Gas Mixtures
Ongoing Research Project, Department of Mechanical Engineering, BUET

- Investigating CO_2 self-diffusivity in MOF pores under co-adsorbed N_2 through molecular dynamics simulations using LAMMPS.
- Comparing tight- and open-pore frameworks to quantify diffusion slowdown relevant to carbon capture and gas-separation applications.

Thermal Conductivity in Doped Stanene/hBN van der Waals Heterobilayers: A Molecular Dynamics Study
Ongoing Research Project, Department of Mechanical Engineering, BUET

- Investigating the effects of substitutional doping on thermal conductivity of stanene/hBN van der Waals heterobilayers using LAMMPS.
- Performing phonon density-of-states (PDOS) analysis to examine changes in spectral overlap and interfacial phonon coupling.

Design and Performance Enhancement of a Double Pipe Heat Exchanger Using Modified Turbulator Fins
Thermo-Fluid System Design (ME 310)

- Designed a modified double-pipe heat exchanger with turbulator fins and evaluated thermal performance and pressure drop using HTRI Xchanger Suite and SolidWorks Flow Simulation.
- Fabricated and tested a working prototype, including leak and pressure testing, with the mechanical design developed in SolidWorks.

Comparative Analysis of a Two-Wheeled Self-Balancing Robot Using P, PI, and PID Controllers
Control Engineering (ME 461)

- Modeled the nonlinear inverted-pendulum system and designed P, PI, and PID controllers using MATLAB, Simulink, and Python with classical and metaheuristic tuning methods.
- Achieved improved transient response using a GA-tuned PID controller with less than 1% overshoot and undershoot.

Stress Analysis of a Double Wishbone Suspension System
Machine Design Sessional (ME 352)

- Modeled the suspension in SolidWorks and performed ANSYS structural analysis under combined braking, cornering, and vertical loads, validating the results using analytical calculations.

Interplanetar (BUET Mars Rover Team)
Mechanical Subteam Lead & R&D Lead

- Led R&D and redesign of rover mechanical systems to improve performance and reliability.
- Re-engineered mechanical subsystems around locally available components with consideration of manufacturability.

Automaestro (BUET Formula Student Team)
Manufacturing Lead

- Designed vehicle components around locally available automotive parts and coordinated fabrication across local workshops.

PROFESSIONAL EXPERIENCE

Junior Mechanical Engineer **June 2026–Present**
Installation of Locally Developed Traffic Signal Project, Department of Civil Engineering, BUET

- Design mechanical components, mounting systems, and IP65 enclosures for traffic signal controllers and associated electronic equipment, considering manufacturability, assembly, environmental protection, and electrical integration.
- Develop fabrication-ready CAD designs for sheet metal manufacturing, 3D printing, and injection molding, and coordinate prototyping and fabrication with multidisciplinary engineering teams.

TRAINING AND CERTIFICATION

- **Industrial Training – Eastern Refinery Limited, Chattogram** (2025)
- **COMSOL Multiphysics Simulation of Thermofluidic Problems** – DCE, BUET (2024)
- **Certified SolidWorks Associate (CSWA)** – Dassault Systèmes (2023)
- **MATLAB Onramp & Simulink Onramp** – MathWorks (2023)

ACADEMIC, TECHNICAL & LEADERSHIP ACTIVITIES

- **Mechanical Subteam Lead,** Interplanetar (BUET Mars Rover Team)
- **Manufacturing Subteam Lead and R&D lead,** Automaestro (BUET Formula Student Team)
- **Academic Team Member,** Bangladesh Olympiad on Astronomy and Astrophysics
- **Vice President,** BUET Math Club & BUET Automobile Club

TECHNICAL SKILLS

- **CAD & Design:** SolidWorks
- **Simulation & Analysis:** ANSYS Mechanical, COMSOL Multiphysics, HTRI Xchanger Suite, SolidWorks Flow Simulation, LAMMPS
- **Molecular Simulation & Visualization:** OVITO, VESTA, Packmol, Moltemplate
- **Programming & Control:** MATLAB, Simulink, Python, C++
- **Manufacturing & Prototyping:** Sheet metal fabrication, 3D printing, injection molding, machining/workshop processes, design for manufacturability

AWARDS & HONORS

- **Dean's List,** Bangladesh University of Engineering and Technology (All Semesters)
- **Bronze Medal,** Global e-Competition in Astronomy and Astrophysics (GeCAA), 2020
- **Champion, Mechathon,** Mechanical Festival, Mechanical Engineering Association, BUET (2026)
- **Most Meritorious Student of the Year,** Srijonshil Medha Onneshon (Dhaka Metropolitan Education Region, Govt. of Bangladesh), 2017
- **National Olympiad Placements:** Champion (Junior Science, 2017), 2nd Runner-Up (Math, 2016), and 21st Place (Physics, 2017) in National Rounds

PERSONAL INFORMATION

- **Father's Name :** Md. Shawkat Anowar Chowdhury
- **Mother's Name :** Meharun Nahar
- **Date of Birth :** 2nd March 2003
- **Nationality :** Bangladeshi
- **Religion :** Islam

REFERENCES

Dr. Md. Ashiqur Rahman
Professor, Department of Mechanical Engineering, BUET

- Phone: 01712274887
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Dr. Kazi Arafat Rahman
Associate Professor, Department of Mechanical Engineering, BUET

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Executive Vice President (Production), Kohinoor Chemical, Tejgaon, Dhaka

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Date: ________________

Md. Arman Hassan